

AKSHAT PANDEY

📞 +91 8770675685 | 🌐 [Portfolio](#) | ✉ akshat16pandey@gmail.com | [in LinkedIn](#) | [Github](#)

SUMMARY

Backend / Python engineer building async, real-time services in **Django** and **FastAPI** (**Celery**, **RabbitMQ**, **PostgreSQL**, **Redis**) — from RTSP/WebRTC video streaming to an internal MLOps pipeline. KAVACH'23 national hackathon winner; B.Tech CS, CPI 9.68.

TECHNICAL SKILLS

Languages: Python, Bash

Backend & APIs: Django, FastAPI, React, REST, WebSockets, Server-Sent Events

Data & Storage: PostgreSQL, Redis, SQLite, SQLAlchemy, Alembic, asyncpg, S3

Async & Streaming: Celery, RabbitMQ, ARQ, asyncio, MediaMTX (RTSP/WebRTC)

Infra & Security: Docker, Kubernetes, Helm, Nginx, AWS, Linux, Git, Ansible, JWT, Argon2id, PAM, scapy, NetworkX

PROFESSIONAL EXPERIENCE

Backend Developer

June 2024 – Present

Intozi Tech. Pvt. Ltd.

Gurugram, India

- Own the **Python** backend across the company's computer-vision and video-analytics products — including the core analytics product **Ikshana** — coordinating frontend work and feature delivery within a small team.
- Build the backend for a client-facing **Video Management System (VMS)** in **Django**, integrating **MediaMTX** for RTSP/WebRTC streaming so live camera feeds and AI analytics surface in the app in real time.
- Architect the platform's data and async layers with **PostgreSQL**, **Redis**, **RabbitMQ**, and **Celery**, offloading video-processing and AI workloads from the request path to keep services responsive and scalable.
- Built an internal **MLOps pipeline** (Django + React) delivering an end-to-end workflow for dataset upload, auto-labeling, human verification, and model training/retraining.

PROJECTS

HeadTogether 🗨️ | *FastAPI, React, WebSockets, Redis, SQLAlchemy*

2026

- Built a social app around **geo-bounded rooms** — ephemeral chat spaces discoverable only by users within a configurable radius, with distance-ranked discovery and geo-privacy that rounds coordinates for non-members.
- Engineered the backend across **73 REST routes** and 20 ORM tables (layered routes/services/repositories) with real-time chat over **WebSockets** and a Redis pub/sub fanout, plus **JWT** auth with refresh-token rotation, Argon2id hashing, and rate limiting.

Papyrus 📄 | *FastAPI, React, Celery, PostgreSQL, Redis, S3, Docker*

2026

- Built a self-hostable, privacy-first PDF web app (merge, split, compress, rotate, reorder, OCR) with a fast browser UI backed by an asynchronous **Celery** pipeline.
- Designed an async **FastAPI** backend (SQLAlchemy 2.x, asyncpg, Alembic) with **S3**-compatible object storage and a zero-retention mode that purges documents on a TTL; containerized the full stack with Docker Compose and Kubernetes/Helm deploy artifacts.

ShieldBuntu 🛡️ | *FastAPI, React, Ansible, SQLite*

2024 – 2026

- Built a one-click Ubuntu hardening tool that turns **CIS Benchmark** controls into **16 idempotent Ansible roles** (SSH, firewall, kernel sysctl, AppArmor, auditd, fail2ban, USB control, and more), each supporting apply, dry-run check, and snapshot-based revert.
- Designed a **FastAPI** daemon that discovers roles and exposes them as a typed HTTP API, streaming every Ansible event live to a **React** UI via **Server-Sent Events**, with a concurrency-safe orchestrator and **PAM**-backed auth with sudoer-group enforcement.

MeshHawk 🦇 | *FastAPI, React, scapy, NetworkX, ARQ, PostgreSQL, Redis*

2023 – 2026

- Built a local-first WiFi mesh detector that ingests **.pcap/.pcapng** captures, reconstructs the 802.11 topology with **scapy** into a directed MAC graph in **NetworkX**, and classifies components as meshes via node-degree heuristics — no data leaves the machine.
- Architected an async **FastAPI** backend with an **ARQ**/Redis worker for the analysis pipeline, **PostgreSQL** (JSONB + GIN-indexed results), live monitor-mode capture, and on-demand printable PDF reports rendering each topology as inline SVG.

EDUCATION

Bhilai Institute of Technology

Bachelor of Technology in Computer Science – CPI: 9.68

Durg, India

Aug. 2020 – May 2024

ACHIEVEMENTS

KAVACH'23: Winner of the inaugural nationwide KAVACH'23 cybersecurity hackathon organized by the Government of India.